

Class XI

Total Contact Hours: 200 (Theory & Practical: 180 ; Remedial & Home Assignment:20)

SEMESTER – I

Course Code: COMS (Theory)

Full Marks: 35

Contact Hours: 100

Unit – 1	Computer System and Organisation	15 Marks	Total 30 Hours
	• Basic Computer Organisation ➤ CPU, Primary Memory (RAM, ROM, Cache), Secondary storage device, I/O devices, units of memory (bit, byte, KB, MB, GB, TB, PB). • Classification of Computers ➤ Super, Mainframe, Mini, PC.		4 hours
	• Concepts of Software ➤ Definition of software, types of software – System Software (Translator: assembler, interpreter, compiler, Loader, Linker, Operating System: Definition and functions, types of OS- Single use, Multiuse, Multiprogramming, Multiprocessing, Time sharing), Application Software (Definition and example), Utility Software, concept of GUI and CUI with examples using LINUX (Basic Commands).		9 hours
	• Number System ➤ Binary, Octal, Decimal, Hexadecimal number system, conversion between number system, Weighted Code (BCD, Binary, 84-2-1 code), non-weighted code (GREY, Excess-3), encoding schemes (ASCII, ISCII, unicode), 1's complement, 2's complement.		7 Hours
	• Boolean Algebra ➤ Postulates, logic gates: NOT, AND, OR, NAND, XOR, XNOR, truth tables, De Morgan theorem, SOP, POS, Simplifications using K-Map and Boolean algebra, logic circuits.		10 Hours
Unit – 2	Programming Fundamentals	10 Marks	Total 25 Hours
	• Concept of Programming ➤ Instruction (Definition, Example), Program (definition, example), Programming Language (concept of high level, low level and assembly language), Procedural and Non-procedural programming, Concept of Structured Programming, Object Oriented Programming		2 Hours
	• Algorithm fundamentals ➤ Definition, characteristic of algorithm, recursive and non-recursive algorithms, representation of algorithm using flowchart, pseudo code, efficiency of algorithm, space complexity, time complexity, asymptotic notation- big O, big Omega, big Theta.		18 Hours
	• Introduction to Problem Solving ➤ Steps for Problem Solving (analysing the problem, developing an algorithm, coding, testing, debugging).		5 Hours

Unit – 3	Introduction to C	10 Marks
	• Basic Structure ➤ Character set, keywords, identifiers, constants, variables and type declaration, Sample programs, pre-processor.	2 Hours
	• Operators ➤ Arithmetic, Relational, Logical, Assignment, Increment and Decrement, Conditional, comma; operator precedence and associativity; arithmetic expression-evaluation and type conversion. Character I/O, Escape sequence and formatted I/O.	3 Hours
	• Branching and Looping ➤ if, if-else, while, do-while, for.	3 Hours
	• Arrays and Structure ➤ One-dimensional and Two-dimensional, Different types of uses. String handling with arrays – read and write, concatenation, comparison, string functions. ➤ Structures: Initialization; arrays of a structure, arrays within structures, structure within structure.	12 Hours
	• User defined functions ➤ Need, Call by Reference, call by value, return value and types, nesting of functions, recursion.	10 Hours
	• Pointers ➤ Declaration and initialization, operators, pointer arithmetic's, accessing variables, pointer & arrays, strings, functions.	15 Hours

SEMESTER – II

Course Code: COMS (Theory)

Full Marks: 35

Contact Hours: 80

Unit – 1	Data Structure	15 Marks	Total 45 Hours
	• Definition, types of data structure-linear and non-linear.		1 Hour
	• Abstract Data types.		1 Hour
	• Arrays: 1D, 2D and their applications.		7 Hours
	• Linked List: Single, circular and double link list.		10 Hours
	• Stack ➤ Stack operations (push and pop), implementation using array and list, application of Stack.		6 Hours
	• Queue ➤ Queue operation implementation using array and list, circular queue, de-queue, priority queue.		6 Hours
	• Recursion ➤ Definition. ➤ Advantages and limitations of recursion. ➤ Understanding what goes behind recursion (internal stack implementation), tail recursion.		4 Hours
	• Searching and Sorting ➤ Linear Search, Binary Search and their comparison. ➤ Bubble Sort and its implementation.		10 Hours
Unit – 2	Computer Networks	10 Marks	Total 20 Hours
	• Introduction to Networking ➤ Analogue and digital Communication. ➤ Mode of Communication- Simplex, half duplex and full duplex. ➤ Network Architecture- Client server, Peer to Peer. ➤ Serial and Parallel Communication. ➤ Measuring Capacity of Communication Media (bandwidth, channel capacity, baud). ➤ Synchronous and asynchronous Transmission Mode. ➤ Baseband and Broadband network.		6 Hours

	• Transmission Media ➤ Wired Communication Media (Twisted Pair, Co-axial cable, Fiber Optic). ➤ Wireless Communication Media (Radio wave, Microwave, Infrared, Satellite).	3 Hours	
	• Network Connecting Devices ➤ Modem, Ethernet Card, RJ45, Repeater, Hub, Switch, Router, Gateway, Wifi card.	2 Hours	
	• Network Type and Topologies ➤ Types of Network-LAN, MAN, WAN. ➤ Network Topologies- Bus, Star, Ring, Tree.	3 Hours	
	• Network Protocols -HTTP, FTP, PPP, SMTP, TCP/IP, POP3, TELNET, HTTPS, VoIP.	2 Hours	
	• Referential Model- OSI Model (Basic Concept, use of devices and protocols at different layers).	1 Hour	
	• Introduction to Web Services: WWW, HTML, XML, IP Addresses, Domain names, URL, ISP, Website, Web browser, Web Server, Web Hosting.	3 Hours	
Unit – 3	Ethics	10 Marks	Total 15 Hours
	• Digital Footprints.		1 Hour
	• Data Protection: Intellectual property rights (copyright, patent, trademark), violation of IPR (plagiarism, copyright infringement, trademark infringement), open-source software and licensing (Creative Commons, GPL and Apache).		5 Hours
	• Cyber Crime: Definition, hacking, eavesdropping, phishing and fraud emails, ransomware, cyber trolls, cyber bullying.		3 Hours
	• Cyber safety: Safely browsing the web, identity protection, confidentiality.		2 Hours
	• Malware: Viruses, trojans, adware.		1 Hour
	• E-waste management: Proper disposal of used electronic gadgets.		2 Hours
	• Information Technology Act: (IT Act).		1 Hour